



```
dataset('iris') from seaborn
```

1. Basic Data Exploration

DSA103 Advanced Chemical Data Science

Lecture 8: Some more EDA & Finding data and deciding what it's good for

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```
int(iris.head())
```

Summary of dataset

```
int("\nDataset Info:")
```

```
int(iris.info())
```

Statistical summary

```
print(iris.describe())
```

5	4.6, 3.1, 1.5, 0.2, 0, setosa
6	5.0, 3.6, 1.4, 0.2, 0, setosa
7	5.4, 3.9, 1.7, 0.4, 0, setosa
8	4.6, 3.4, 1.4, 0.3, 0, setosa
9	5.0, 3.4, 1.5, 0.2, 0, setosa
10	4.4, 2.9, 1.4, 0.2, 0, setosa
11	3.1, 1.5, 0.1, 0, setosa
12	3.7, 1.5, 0.2, 0, setosa
13	3.4, 1.6, 0.2, 0, setosa
14	3.0, 1.4, 0.1, 0, setosa
15	3.0, 1.1, 0.1, 0, setosa
16	4.0, 1.2, 0.2, 0, setosa
17	5.7, 4.4, 1.5, 0.4, 0, setosa
18	5.4, 3.9, 1.3, 0.4, 0, setosa
19	5.1, 3.5, 1.4, 0.3, 0, setosa
20	5.7, 3.8, 1.7, 0.3, 0, setosa
21	5.1, 3.8, 1.5, 0.3, 0, setosa
22	5.4, 3.4, 1.7, 0.2, 0, setosa
23	5.1, 3.7, 1.5, 0.4, 0, setosa
24	4.6, 3.6, 1.0, 0.2, 0, setosa
25	5.1, 3.3, 1.7, 0.5, 0, setosa
26	4.8, 3.4, 1.9, 0.2, 0, setosa
27	5.0, 3.0, 1.6, 0.2, 0, setosa
28	5.0, 3.4, 1.6, 0.4, 0, setosa

Content overview

EDA continued:

- Dimensionality reduction by PCA in EDA
- Data Analysis continued: Applying simple ML models

Finding data

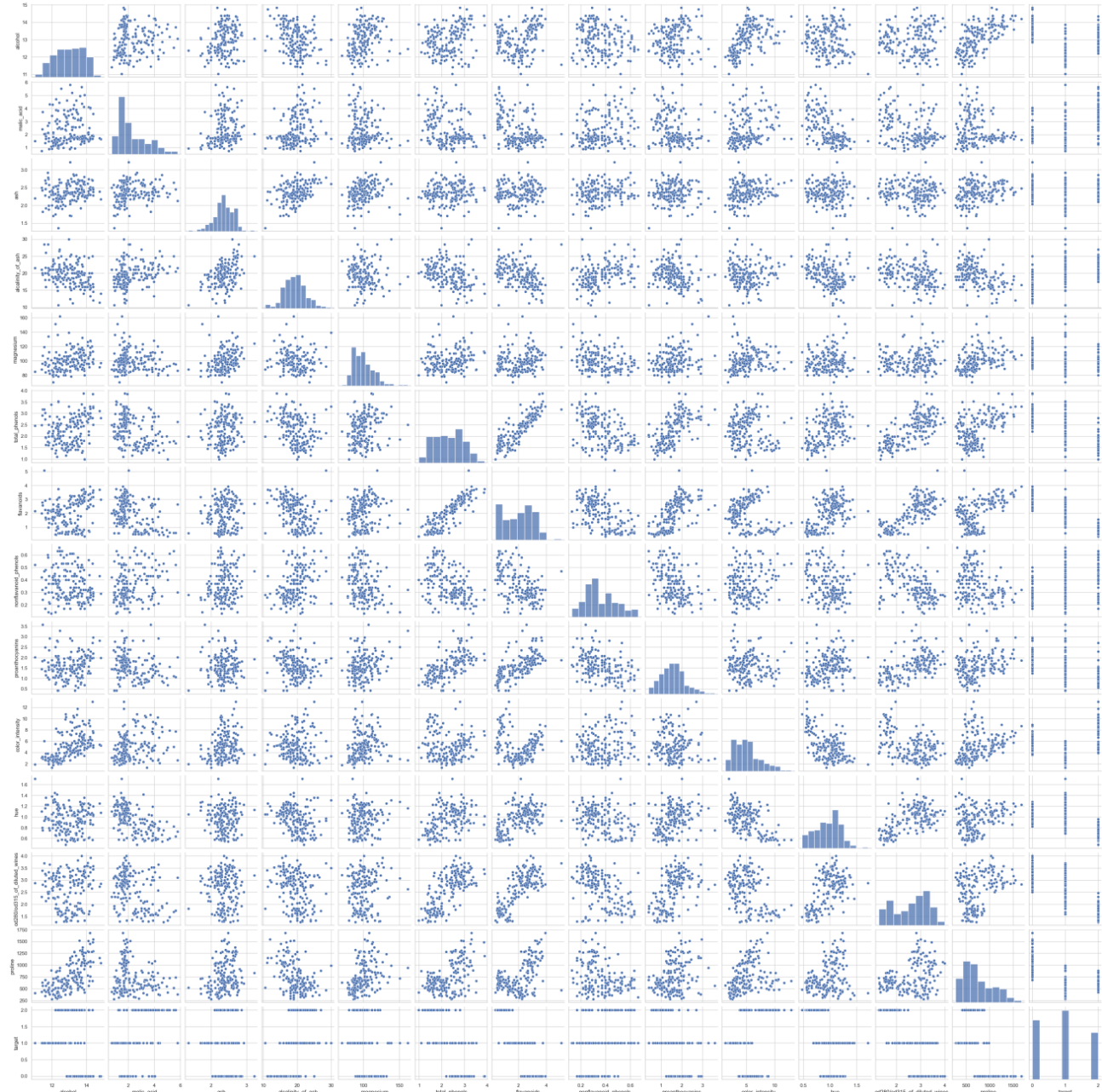
- What, how and where?
- Is the data any good?
- What is the data good for?



Dimensionality reduction – the visualisation problem

High dimensional data has some practical drawbacks:

- **Difficult visualisation**
- Slower training for ML models (more features)
- Overfitting (more noise than patterns)
- Multicollinearity (quite a few features may be correlated)



Dimensionality reduction techniques

Reduce number of features while preserving as much information (**variance**) as possible.

Basically, two types:

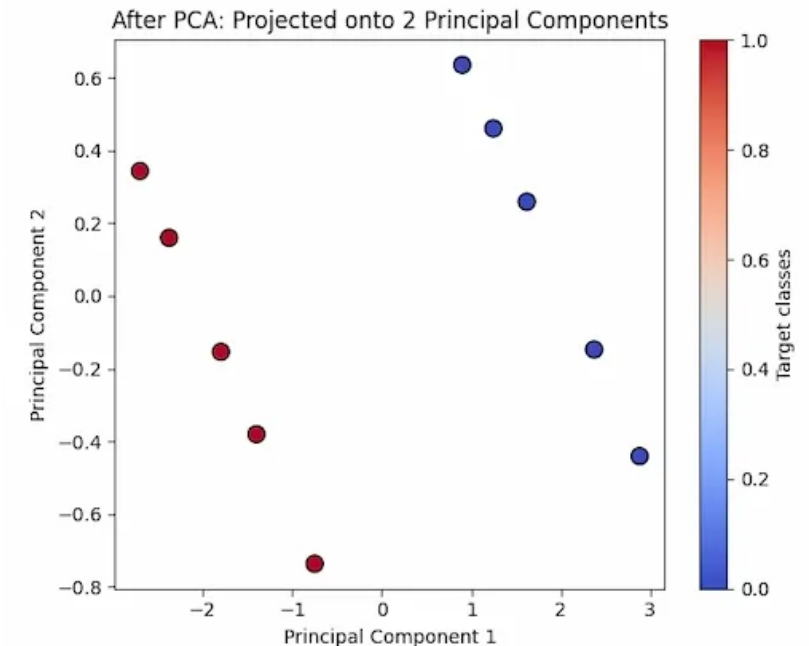
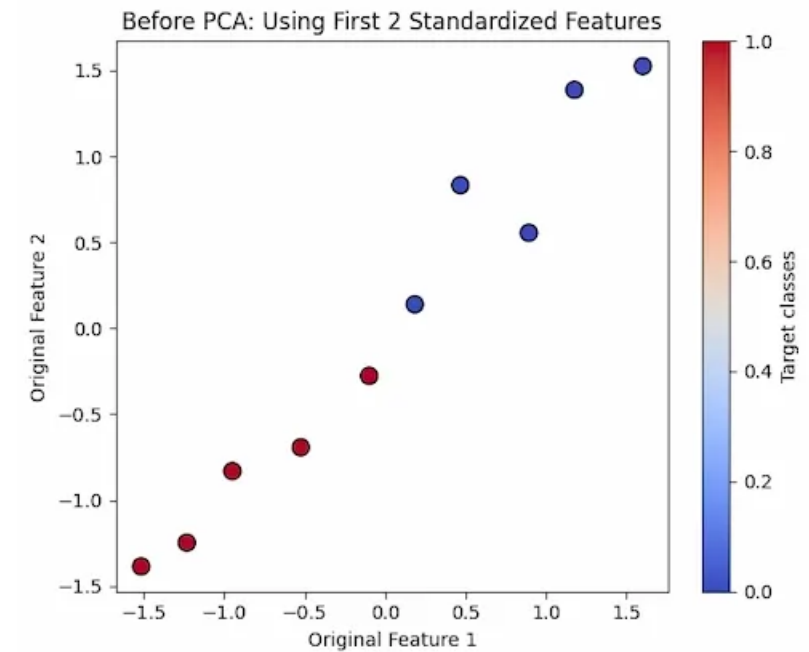
- 1) **Feature selection:** Original features are preserved, but fewer of them will be taken into account (e.g. **Filters**)
- 2) **Feature extraction:** new, compressed representations of the data are obtained by **transforming or combining original features** to capture most of the variance / information in the data. E.g.:
 - **PCA (Principal Component Analysis):** most popular method, linear technique
 - t-SNE (t-distributed Stochastic Neighbor Embedding): non-linear technique
 - Autoencoders: neural network-based

Dimensionality reduction: PCA

Working principle:

- 1) Standardising the data (so each feature has equal importance)
- 2) Calculate covariance matrix to check how features relate to each other. Results in eigenvectors (direction of PC) and eigenvalues (importance)
- 3) Finding directions (“principal components”) where the data varies the most:
 - **1st Principal Component (PC1):** The direction of maximum variance (most spread).
 - **2nd Principal Component (PC2):** The next best direction, *perpendicular to PC1* and so on.
- 4) After finding the components that capture the **maximum variance**, the original data is projected onto these new axes

Graphs from: <https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/>

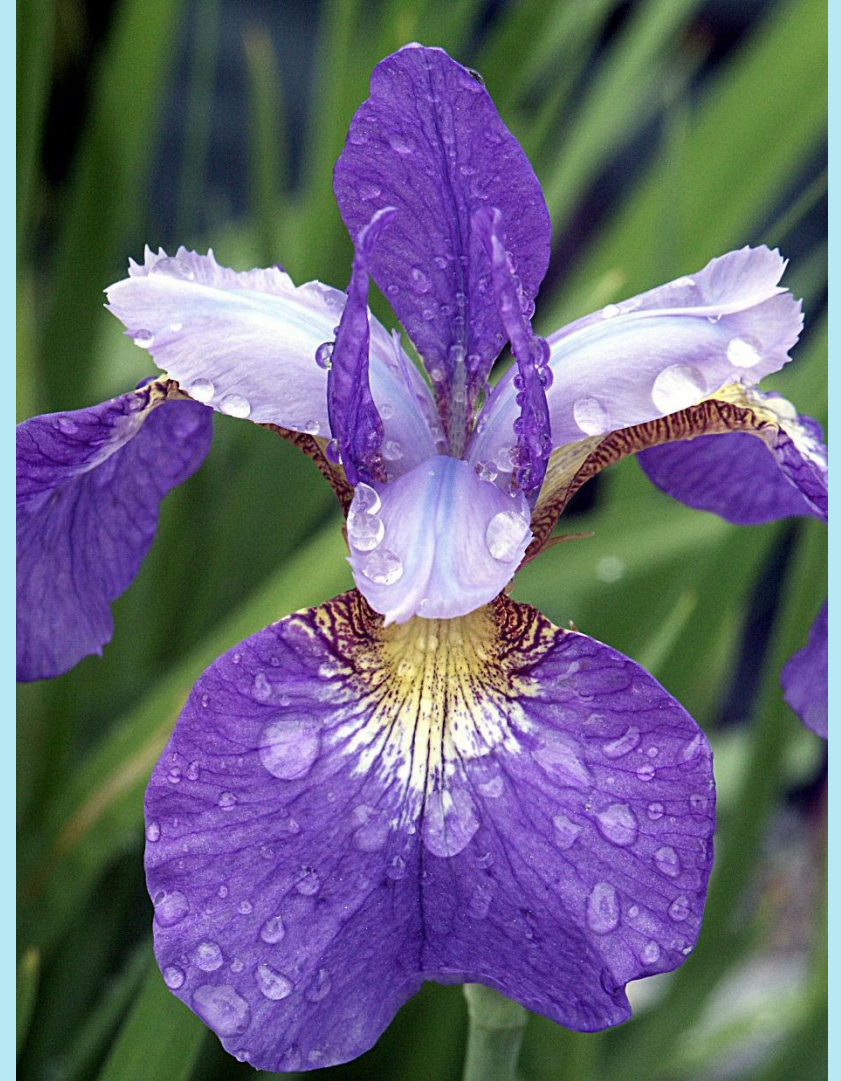


Activity: EDA on iris dataset

Perform a PCA on the iris data set to visualise all four features in a 2D plot.

Compare to the plots of the bivariate analysis and discuss, which features are showing the highest variance (main features of the primary components).

Discuss how to interpret the PCA plot.



EDA continued: A first glance at ML

Machine learning (ML) = models are fit to previously seen data and yield predictions or classifications based on patterns in the data.

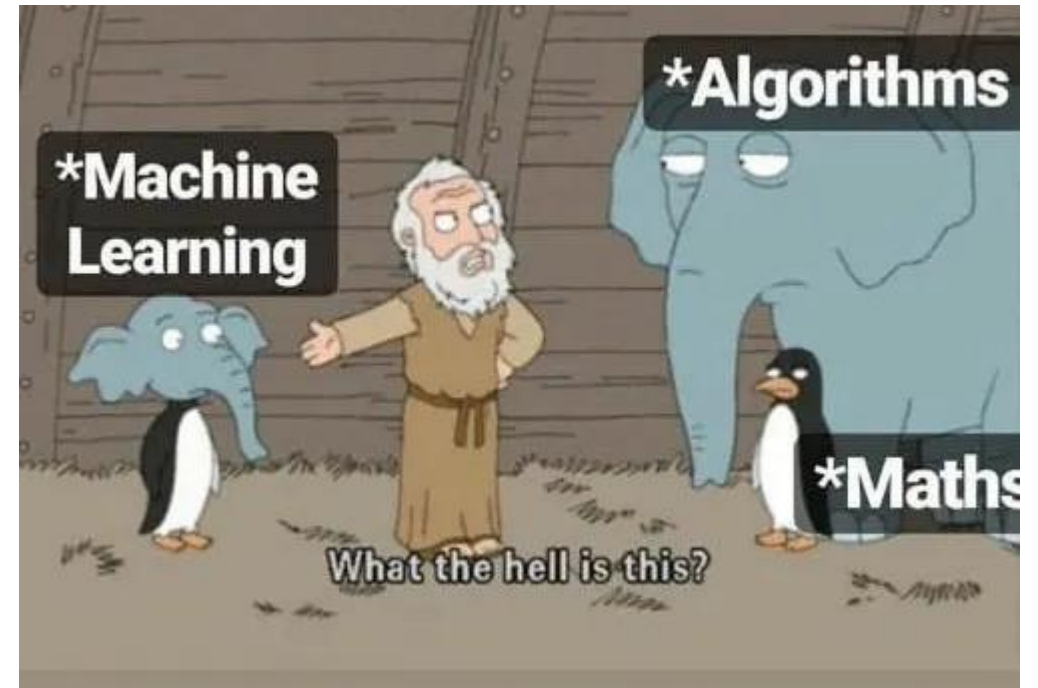
In short:

- Goal: Learn patterns from data automatically.
- Input: Features (measurements, text, pixels, etc.)
- Output: Predictions/classifications (Supervised ML), or patterns (Unsupervised ML)
- Key idea: The computer learns patterns and context in data
- For now: Tool to understand our data better.

We will use models from the **scikit-learn** library:

- PCA
- Logistic regression
- Decision tree

Pic from: <https://medium.com/nybles/understanding-machine-learning-through-memes-4580b67527bf>



How can ML help in data analysis

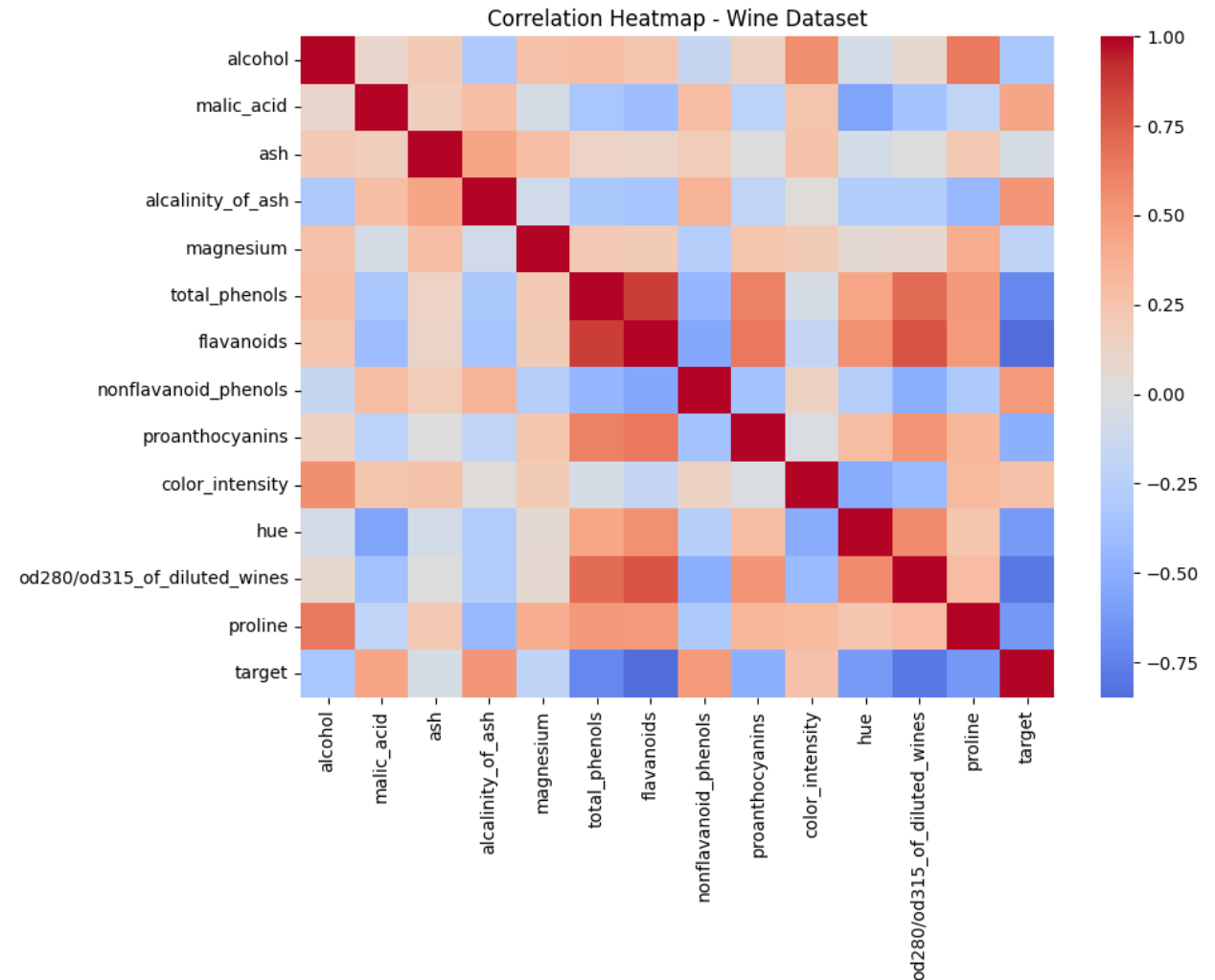
Humans are not good at...

- ...working with unfamiliar categories
- ...recognising complex patterns in multidimensional data

Automated data analysis as (ultimate) solution?

Maybe. But practically, it **can help with**:

- Feature importance
- Detecting nonlinearities
- Predictability baseline
- Finding hidden structures/patterns
- Outlier and anomaly detection
- Spotting data leakage
- Preprocessing



A first glance at ML: Terminology

Feature matrix (X): Features used for learning (2D array, DataFrame)

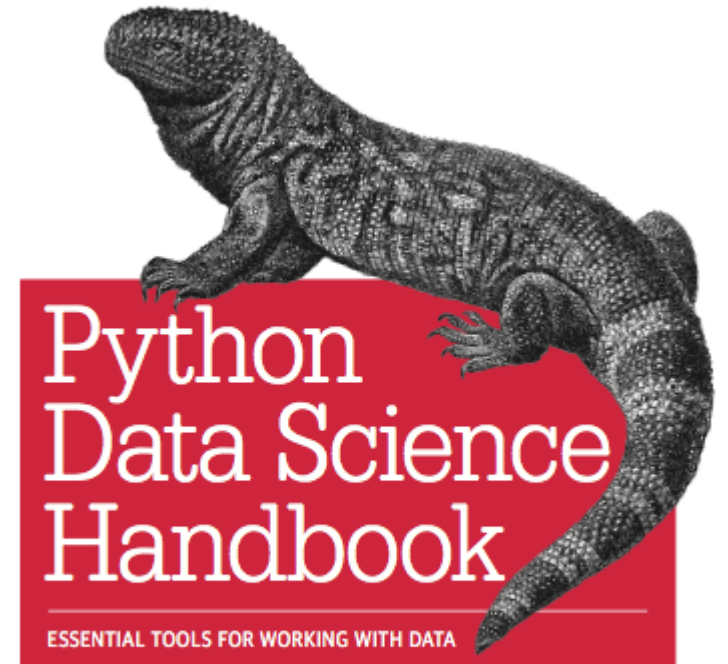
Labels / targets (y): Data to be predicted by ML model (mostly 1D arrays, Series)

Model: Function or set of rules that learns from X and predicts y, or that tries to find patterns in data.

Supervised learning: The model will be given examples of what the answer should be (labelled data)

Unsupervised learning: The model tries to find patterns > noise (unlabelled data)

O'REILLY



Jake VanderPlas

<https://jakevdp.github.io/PythonDataScienceHandbook/05.02-introducing-scikit-learn.html>

A first glance at supervised ML: Scikit-learn Recipe

- 1) Choose a class of model by importing the appropriate estimator class from Scikit-Learn.
- 2) Choose model *hyperparameters*
- 3) Arrange data into a features matrix and target vector
- 4) Fit the model to your data by calling the **fit()** method
- 5) Apply the Model to new data: predict labels for unknown data using the **predict()** method.

<https://jakevdp.github.io/PythonDataScienceHandbook/05.02-introducing-scikit-learn.html>

Activity: Introducing ML on the iris dataset

Applying Logistic Regression and a Decision tree as EDA tools

Assignment for Session 8: Applying EDA for a slightly bigger dataset.

Finding data – what?

You will find data for almost everything you look for...

...and normally, you look for data with a purpose in mind.

What types of data do we generate in a chem or bio lab? What challenges do we expect for these categories?

Type	Example	Examples of databases
Molecular structures	Small molecules, proteins	RSCB PDB , CCDC/CSD , PubChem
Spectroscopy data	NMR, MS, IR	SDBS , nmrshiftdb2
Genomic/proteomic	Sequences, expression	GenBank , UniProt
Reaction data	Synthetic pathways, large DS on reaction optimization	Scifinder , Reaxys , Open reaction database
Computational data	Thermochemical data for small molecules, computed structures	CCCBDB , Zinc

Activity: Finding data

Demo:

- 1) Searching for a molecule in [PubChem](#)
- 2) Finding a protein in [RSCB PDB](#)
- 3) Searching for reaction data in [Scifinder](#)

Exercise:

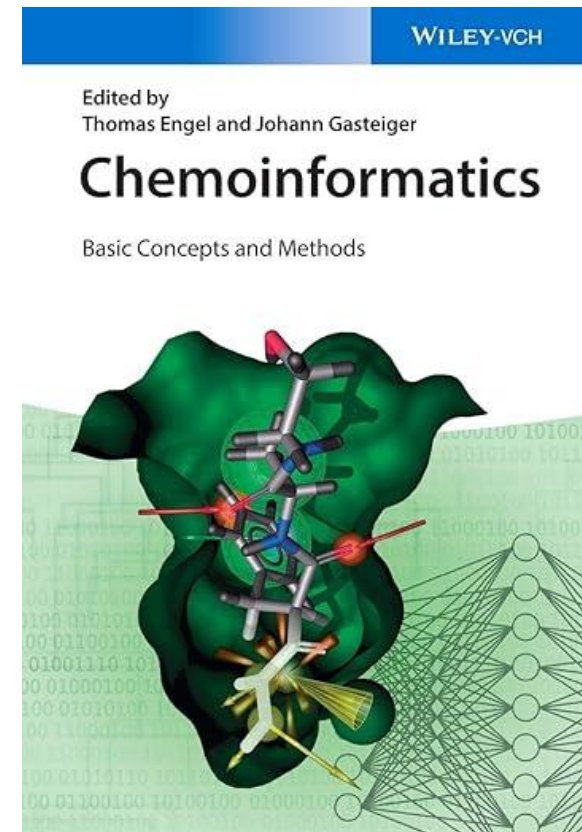
- Use case: find a database entry for caffeine (try e.g. [PubChem](#), [ChemSpider](#), [CRC Handbook](#), [SDBS](#), [nmrshiftdb2](#), [ChEMBL](#))
- Investigate the metadata. Actually, what is metadata?
- What type of data is available?
- What can you learn from the entry?
- What data is available, which information is missing?
- Discuss your findings in pairs or small groups

Finding data – how and where?

There is plenty of resources out there, just need to find them 😊

Some starting points to find data:

- 1) Database systems / search portals:
 - Scifinder, Reaxys: most important database systems for chemical information
 - NCBI: Biotechnology information platform linking to most important specific databases
 - Expasy: Swiss bioinformatics resource portal
- 2) (Public) Databases for specific purposes:
 - PDB, geneontology, Ensembl (genomes)
- 3) (Via literature) In repositories
- 4) In literature (e.g. use bibliographic databases, e.g. C.A.B. International via Scifinder, Web of Science)
- 5) In analogue documentation (use indexing systems or expert knowledge)



Zass, E. Engel, T (2018) *Databases and Data Sources in Chemistry* In Engel, T., Gasteiger, J., *Chemoinformatics* (1st ed., pp. 185 – 230), Wiley-VCH.

Finding data – where?

General data - for training purposes and your own interest:

- Kaggle: <https://www.kaggle.com/datasets>
- Federal Statistical Office: <https://www.bfs.admin.ch/bfs/en/home/statistics/catalogues-databases/data.html>
- OpenData: <https://opendata.swiss/en/>
- OpenData Zurich: <https://www.stadt-zuerich.ch/opendata.secure.html>
- United Nations: <http://data.un.org/>
- World Health Organization: <http://apps.who.int/gho/data/node.home>
- World Bank: <https://data.worldbank.org/>
- Cern: <http://opendata.cern.ch/>
- Nasa: <https://data.nasa.gov/>
- FiveThirtyEight: <https://github.com/fivethirtyeight/data>

~~Is the data any good?~~ Better: Is the data fit for my purpose?

Universal quality indicators, e.g.:

- **Provenance / Source:** Who collected the data, when, under what conditions
- **Metadata completeness:** Are conditions, methods, instruments, and identifiers fully described?
- **Standardization / Format:** Are chemical identifiers and units standardized (e.g., InChI, SMILES, μM vs mM)?
- **Reproducibility / Replicates:** Information on replicate measurements, or standard deviations?
- **License & Accessibility:** Can you legally reuse or redistribute the data?
- **Documentation / Curation level:** Human curation or automatic upload?
- **Versioning:** Can you trace updates or corrections?

Domain specific quality indicators, e.g.:

- Spectroscopy: signal-to-noise ratio (S/N), baseline correction, info on instrument calibration, mass accuracy, peak assignment confidence, sample preparation details, ...
- Chemical reactions: yield reporting, full details on reaction conditions, source reliability, identifiers, ...

Deciding what the data is good for

Points to consider:

- What does the data actually present?
- What is the scope and scale of the data?
- How was the data produced?
- What metadata is available?
- What kind of questions can the data answer?
- What about limitations and biases?
- Is the data trustworthy?
- Can this data be combined with other data sources?
- What about ethical or legal issues?
- What story could the data tell?

Activity: Assessing data quality and what we can do with the data

Let's look again at an entry in PubChem:

- What does the data show?
- Where does the data come from?

Now imagine a collection of data:

- What can (and can't) we do with it?